Predicting Employee Retention at Springboard

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Executive Summary

- Springboard is a high-growth startup for which employee turnover can be costly and disruptive.
- In this report, we use Employee Pulse Survey data to identify the features most predictive of employee stated intent to stay at Springboard for one year.
- We find that Employee Net Promoter Score (ENPS), the length and sentiment of employee responses to the question "If your score (on ENPS) is anything between 0-8, what would it take to move your rating to 9 or 10?" and a few cultural factors such as a sense of belonging, enjoying the work, and feeling comfortable giving feedback are most strongly predictive of employee intent to stay.
- Our models are particularly predictive for employees who have indicated they intend to leave Springboard within one year.
- Given our findings, we recommend the following:
    - **Springboard should run non-anonymous employee pulse surveys (without asking about intent to stay to avoid biasing the responses) asking for a few of our key predictive features** such as ENPS quarterly. Springboard can then use our model to target employees who are predicted to leave for intervention.
    - **Springboard should focus on cultural factors** such as a sense of belonging and its environment for giving feedback to address employee attrition.
    - **Springboard should ensure question stability between employee pulse surveys.** This will allow for more powerful statistical models to be built over time.

* We are okay sharing this project with future students.

Section I: Introduction

Founded in 2013, Springboard is an online learning platform that helps train students to become software engineers, data scientists, and other tech professionals. Like many growing companies, employee turnover can be particularly disruptive for Springboard. In this report, we use Springboard's employee pulse survey data to identify the attributes of employees who are most likely to say in the survey that they either "agree" or "strongly agree" with the statement "I see myself working at Springboard a year from now." By identifying the features most strongly predictive of stated intent to stay, we believe we can help Springboard identify the employees most at risk of leaving so that Springboard can take action and reduce employee attrition.

Section II: Data

For our analysis, we relied on three sources of data. Our primary source of data is **Springboard's Spring 2022 Employee Pulse Survey**, which was provided by Springboard and contains 180 anonymous survey responses to a variety of questions, ranging from "I feel comfortable giving feedback to my colleagues" (answer is on a 5-point scale from Strongly Disagree to Strongly Agree) to "Do you have any additional feedback to share about your organization?" (answer is free response text). Importantly, the survey asks employees to respond to the prompt "I see myself working at Springboard a year from now" with a response on a 5-point scale from Strongly Disagree to Strongly Agree. This "intent to stay" is the response variable that we predict in our analysis below.

See **Exhibit 1** for a snapshot of the May 2022 Employee Pulse data, which contains 180 observations and 48 features. As this was the initial data provided to us by Springboard, we performed the most in-depth analysis on this survey. In addition to the original 2022 survey, after we had performed some initial analysis, we were able to obtain two additional pulse survey datasets. One is from February 2023 and contains 243 responses while the other is from January 2022 and contains 161 responses. While the question set across the 3 pulse surveys is not stable, all 3 surveys do include our "intent to stay" response variable and contain a number of overlapping questions. Therefore, as described below, we supplement our analysis of the May 2022 survey with an analysis of the 3 combined surveys.

Section III: Methods

Data Cleaning – Given the varied nature of the survey response fields, some of which asked participants to respond on a 5-point scale of Strongly Disagree / Disagree / Neutral / Agree / Strongly Agree, some of which asked participants to choose from a dropdown menu, and some of which asked for open text free responses, we first divided the data into three categories.

We called the 5-point scale questions **"numerical"** and translated those responses to a scale from 1-5 where "Strongly Agree" was a 5 and "Strongly Disagree" was a 1. For a few of the dropdown menu questions where there was clearly a distinct ranking (e.g., "the overall review cycle timelines were" a) "too long", b) "just right", or c) "too short") we translated that ranking into a numerical scale (in the example above, a scale from 1-3). We called the remaining dropdown menu questions **"categorical"** (e.g., "Which team are you on?") and created a series of dummy variables ("one-hot" encoding) for each response category. Finally, we identified 4 questions that were open text free response (e.g., "What changes would make you feel better about Springboard's approach to hybrid work") which we called **"free form"** questions, and which could not be translated to numerical or dummy variables. In our initial models, we treated the number of characters in these free form responses as a numerical variable.

Sentiment Analysis – Given treating the length of the free form questions as a numerical variable is likely not an ideal approach, we also employed sentiment analysis on these 4 free form questions to make our predictive models more powerful. Our sentiment analysis took 2 approaches. First, we ran these 4 questions through Hugging Face's Bidirectional Encoder Representations from Transformers (BERT) trained on Stanford AI's Large Movie Review Dataset. This preprocessing step translated each of the 4 free form response columns into a numerical column on a scale from 1-5.

Second, we took a pre-trained BERT model and stripped off the last layer, so that each free form response was distilled into 768 sentiment features. We then used Principal Component Analysis (PCA) to map those 768 features down to 5 sentiment features per free form response column. This approach yielded 20 sentiment features in total (5 PCA columns x 4 free form columns).

Our final May 2022 cleaned dataset was 180 rows x 114 features excluding sentiment, 180 rows x 118 features with the BERT sentiment scale, and 180 rows x 138 features with BERT and BERT PCA.

Model 1: May 2022 Data Numerical Features[1] – Given the high ratio of features to observations as well as the typically strong performance of models such as random forests, GBM, and XGBoost which already do a form of automatic feature selection (and in some cases regularization) by choosing the best splits within component trees, we decided to rely on the models to do automatic feature selection rather than doing manual feature selection. Specifically, for our initial model, we used the May 2022 dataset without sentiment analysis and performed 5-fold stratified (to preserve the percentage of each class) cross-validation on 5 multiclass prediction models: Logistic Regression with Lasso (where the shrinkage parameter was tuned via

[1] Here, "Model 1" is used in the collective sense; it refers to the complete set of models (Lasso, Random Forest, GBM, CatBoost) we trained using the original dataset. Similarly, "Model 2" refers to the same set of models, but trained on the original dataset with sentiment analysis features. The same terminology applies to "Model 3".

5-fold cross-validation), Random Forest, Gradient Boosted Machines (GBM), XGBoost, and CatBoost.

We formulated these models as a multiclass classification problem where the features were used to predict the probability that someone would respond "Strongly Agree", "Agree", "Neutral", "Disagree", or "Strongly Disagree" to the question "I see myself working at Springboard a year from now". To evaluate the models as predictors, we then determined that their "prediction" would be the class with the highest predicted probability and then bucketed their predictions into two classes: 0 if the predicted response was "Strongly Disagree", "Disagree", or "Neutral" and 1 if the predicted response was "Agree" or "Strongly Agree" (the employee intends to stay at Springboard). We used multiclass classification and then mapped back to two class classification rather than two class classification from the start because we believe that, as Springboard collects more data, they can potentially derive more value from targeting "Strongly Disagree" respondents for action and therefore identifying "Strongly Disagree" respondents versus "Disagree" or "Neutral" respondents is of value.

Exhibit 2 shows the results of our Model 1 cross-validation. **Random Forests** performed the best with an average AUC of **0.727**.

Model 2: May 2022 Data with Sentiment Analysis – We extended Model 1 by performing the same 5-fold stratified cross-validation on Logistic Regression with Lasso, Random Forests, GBM, XGBoost, and CatBoost on the May 2022 survey data incorporating first our simple sentiment analysis and second our PCA sentiment analysis. As shown in **Exhibit 3**, our best-performing model was a **Random Forest** model incorporating both the simple sentiment scale as well as the PCA sentiment data. This model delivered an average AUC of **0.747**, which is a significant improvement over our simple Model 1.

Model 3: Three Survey Model – Finally, we ran 5-fold stratified cross-validation on Logistic Regression with Lasso, Random Forest, GBM, XGBoost, and CatBoost on the dataset which incorporated the **January 2022, May 2022, and February 2023** survey data. This allowed us to determine the tradeoff between having more observations (395 observations versus 180 originally) and having fewer predictors (80 predictors versus 138 in our final Model 2). As shown in **Exhibit 4**, the larger model performed similarly to our Model 2, with **Random Forests** delivering an average AUC of **0.728** .

Section IV: Results

As described above and shown in **Exhibits 2, 3, and 4**, our two best models were a Random Forest model on the May 2022 survey data incorporating sentiment analysis and a Random Forest model on the full dataset, both of which delivered AUCs of ~0.73-0.75.

More interestingly for Springboard, **Exhibits 5 and 6** show the feature importance for Models 2 and 3. Perhaps unsurprisingly, the most predictive feature is the response to the question "On a scale of 1-10, how likely are you to recommend Springboard to a friend as a place to work?", which is part of the traditionally-accepted metric of **Employer Net Promoter Score (ENPS)**. Similarly, the other top predictors outside of ENPS and features related to the free response variable associated with ENPS are a sense of belonging at Springboard, a sense that one's workload is reasonable, enjoying the work, and feeling comfortable giving one's manager feedback. Given these features are similarly important across models (which are trained on slightly different datasets), we believe that they are generalizable outside of just the 2022 and 2023 surveys.

We also evaluated each model's true positive rate, false positive rate, and precision by predicted percentile of intent to stay (which corresponds nicely to the 1-5 scale for intent to stay in the survey). As shown in **Exhibit 7**, almost all of our models (these are the various specifications of Model 2), perform extremely well at predicting true positives amongst employees who are likely to say they strongly disagree or disagree that they will be at Springboard in one year. We believe that this shows that our model can be an extremely useful tool for Springboard to identify the employees who are likely to leave (as described below) and target those employees with an intervention.

Section V: Conclusion / Recommendations

Given their relatively high AUC scores, we believe that our Models 2 and 3 are good predictors of an employee's intent to stay at Springboard. Given our findings, we would make the following recommendations to improve employee retention:

1. **Focus on improving the underlying conditions that would make an employee respond positively to our most predictive questions.** For instance, our second most important feature in Model 3 is a sense of belonging at Springboard, so focusing on events and culture that foster a sense of belonging is likely important.
2. **Potentially run a lightweight version of the pulse survey that asks our top 10 most predictive questions (excluding intent to stay to avoid bias in the results) that is not anonymous on a quarterly basis.** This might help Springboard use our model to identify the employees most likely to leave, who Springboard could then take action with to try to retain.
3. **In future anonymous pulse surveys that do ask about intent to stay, keep the set of questions stable** so that we can leverage all surveys in the future to make our models more statistically predictive.

Exhibits

Exhibit 1: Snapshot of May 2022 Pulse Survey

Timestamp	I understand how my day to day activities fit into Springboard's strategy and contribute to the mission	Mark the one Springboard value you think... [We, as a company, exhibit most consistently]	Mark the one Springboard value you think... [We, as a company, exhibit least consistently]	Mark the one Springboard value you think... [Our execs exhibit most consistently]	Mark the one Springboard value you think... [Our execs exhibit least consistently]	Generally, I believe my workload is reasonable for my role	I believe I can take time off & disconnect	I see myself working at Springboard a year from now	I enjoy the majority of what makes up my day to day work	My strengths are utilised within my team	I know what constitutes good performance in my role	I am empowered to make decisions within my role	I am learning & growing my skillset in my current role
5/16/22	Disagree	Decisive & execution oriented	Mission First	Decisive & execution oriented	Strive for excellence	Disagree	Disagree	Strongly disagree	Strongly disagree	Strongly disagree	Strongly agree	Neutral	Strongly disagree
5/24/22	Agree	Strive for excellence	Mission First	Always be learning	Mission First	Disagree	Strongly disagree	Strongly disagree	Strongly disagree	Strongly disagree	Strongly disagree	Strongly disagree	Strongly disagree
5/20/22	Agree	Embrace humanness, relationships & fun	Always be learning	Embrace humanness, relationships & fun	Always be learning	Neutral	Disagree	Neutral	Neutral	Disagree	Neutral	Neutral	Disagree
5/16/22	Neutral	Mission First	Open, honest & constructive	Mission First	Take ownership	Disagree	Strongly disagree	Neutral	Neutral	Disagree	Neutral	Neutral	Agree
5/17/22	Strongly disagree	Embrace humanness, relationships & fun	Take ownership	Strive for excellence	Decisive & execution oriented	Strongly agree	Strongly agree	Strongly disagree	Agree	Strongly agree	Strongly agree	Strongly agree	Strongly agree
5/17/22	Neutral	Take ownership	Embrace humanness, relationships & fun	Take ownership	Embrace humanness, relationships & fun	Neutral	Neutral	Agree	Neutral	Neutral	Neutral	Agree	Neutral
5/21/22	Agree	Always be learning	Open, honest & constructive	Decisive & execution oriented	Open, honest & constructive	Neutral	Neutral	Strongly disagree	Neutral	Neutral	Strongly agree	Agree	Neutral
5/16/22	Strongly Agree	Always be learning	Take ownership	Embrace humanness, relationships & fun	Take ownership	Disagree	Disagree	Agree	Agree	Agree	Strongly disagree	Neutral	Agree
5/16/22	Agree	Always be learning	Strive for excellence	Strive for excellence	Mission First	Neutral	Neutral	Neutral	Agree	Agree	Neutral	Neutral	Agree
5/16/22	Agree	Mission First	Decisive & execution oriented	Always be learning	Decisive & execution oriented	Agree	Agree	Neutral	Agree	Neutral	Neutral	Agree	Agree
5/16/22	Agree	Strive for excellence	Decisive & execution oriented	Strive for excellence	Decisive & execution oriented	Agree	Agree	Agree	Agree	Strongly disagree	Strongly agree	Strongly agree	Agree
5/16/22	Agree	Mission First	Strive for excellence	Mission First	Embrace humanness, relationships & fun	Disagree	Strongly disagree	Neutral	Neutral	Neutral	Agree	Strongly agree	Agree
5/17/22	Agree	Embrace humanness, relationships & fun	Decisive & execution oriented	Strive for excellence	Open, honest & constructive	Neutral	Agree	Neutral	Agree	Agree	Agree	Strongly agree	Neutral
5/17/22	Strongly agree	Always be learning	Decisive & execution oriented	Strive for excellence	Decisive & execution oriented	Agree	Neutral	Disagree	Neutral	Agree	Strongly agree	Strongly agree	Agree
5/17/22	Agree	Always be learning	Take ownership	Mission First	Open, honest & constructive	Agree	Agree	Agree	Neutral	Neutral	Agree	Agree	Agree
5/17/22	Agree	Mission First	Take ownership	Mission First	Strive for excellence	Strongly disagree	Disagree	Strongly disagree	Disagree	Agree	Neutral	Agree	Neutral
5/17/22	Agree	Take ownership	Open, honest & constructive	Mission First	Mission First	Neutral	Agree	Neutral	Neutral	Strongly agree	Agree	Agree	Disagree
5/18/22	Strongly agree	Embrace humanness, relationships & fun	Decisive & execution oriented	Embrace humanness, relationships & fun	Open, honest & constructive	Agree	Neutral	Agree	Agree	Agree	Neutral	Agree	Agree
5/19/22	Agree	Embrace humanness, relationships & fun	Strive for excellence	Decisive & execution oriented	Open, honest & constructive	Neutral	Agree	Neutral	Agree	Agree	Strongly disagree	Neutral	Strongly disagree
5/20/22	Strongly agree	Decisive & execution oriented	Strive for excellence	Embrace humanness, relationships & fun	Strive for excellence	Agree	Agree	Strongly agree	Agree	Agree	Strongly agree	Agree	Agree

Note: Response column is highlighted in green. We mapped each of the values (Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree) to a numeric value from 1-5, with 1 corresponding to Strongly Disagree and 5 corresponding to Strongly Agree.

Exhibit 2: Model 1 (May 2022 Data without Sentiment Analysis) Cross-Validation Performance

	LASSO_AUC	RF_AUC	GBM_AUC	XGB_AUC	CAT_AUC
0	0.653846	0.725753	0.687291	0.704013	0.687291
1	0.593645	0.780936	0.682274	0.698997	0.698997
2	0.583333	0.791667	0.708333	0.708333	0.687500
3	0.750000	0.625000	0.625000	0.583333	0.666667
4	0.616364	0.712727	0.692727	0.556364	0.621818

```
Lasso avg AUC using Max Prediction Method = 0.6394  and SE(AUC) = 0.027
   RF avg AUC using Max Prediction Method = 0.7272  and SE(AUC) = 0.0266
  GBM avg AUC using Max Prediction Method = 0.6791  and SE(AUC) = 0.0127
  XGB avg AUC using Max Prediction Method = 0.6502  and SE(AUC) = 0.0296
  CAT avg AUC using Max Prediction Method = 0.6725  and SE(AUC) = 0.0122
```

Exhibit 3: Model 2 (May 2022 with Sentiment and PCA Analysis) Cross-Validation Performance

	LASSO_AUC	RF_AUC	GBM_AUC	XGB_AUC	CAT_AUC
0	0.610368	0.627090	0.704013	0.622074	0.682274
1	0.769231	0.780936	0.545151	0.638796	0.737458
2	0.687500	0.708333	0.708333	0.604167	0.729167
3	0.541667	0.770833	0.625000	0.625000	0.770833
4	0.661818	0.849091	0.581818	0.556364	0.601818

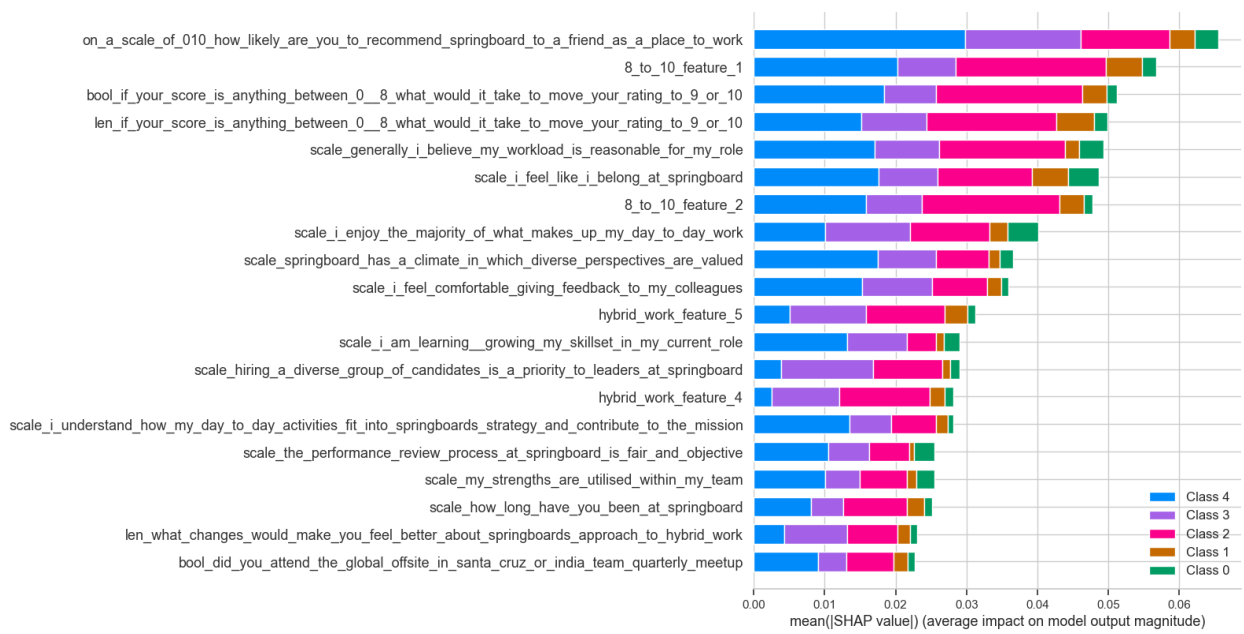
```
Lasso avg AUC using Max Prediction Method = 0.6541  and SE(AUC) = 0.034
   RF avg AUC using Max Prediction Method = 0.7473  and SE(AUC) = 0.0335
  GBM avg AUC using Max Prediction Method = 0.6329  and SE(AUC) = 0.0291
  XGB avg AUC using Max Prediction Method = 0.6093  and SE(AUC) = 0.0128
  CAT avg AUC using Max Prediction Method = 0.7043  and SE(AUC) = 0.0262
```

Exhibit 4: Model 3 (Three Survey Model with Sentiment and PCA Analysis) Cross-Validation Performance

	LASSO_AUC	RF_AUC	GBM_AUC	XGB_AUC	CAT_AUC
0	0.692939	0.755747	0.745074	0.777504	0.803366
1	0.619048	0.705665	0.744663	0.623563	0.606322
2	0.695774	0.777911	0.697368	0.697368	0.795455
3	0.687002	0.723684	0.685008	0.706140	0.692185
4	0.688424	0.677750	0.662562	0.759852	0.727422

```
Lasso avg AUC using Max Prediction Method = 0.6766  and SE(AUC) = 0.013
   RF avg AUC using Max Prediction Method = 0.7282  and SE(AUC) = 0.0159
  GBM avg AUC using Max Prediction Method = 0.7069  and SE(AUC) = 0.0147
  XGB avg AUC using Max Prediction Method = 0.7129  and SE(AUC) = 0.0242
  CAT avg AUC using Max Prediction Method = 0.7249  and SE(AUC) = 0.0324
```

Exhibit 5: Feature Importances for Model 2, Random Forest



Note: Class 0, 1, 2, 3, and 4 correspond to 1, 2, 3, 4, and 5 on the 5-point scale we used.

Exhibit 6: Feature Importances for Model 3, Random Forest

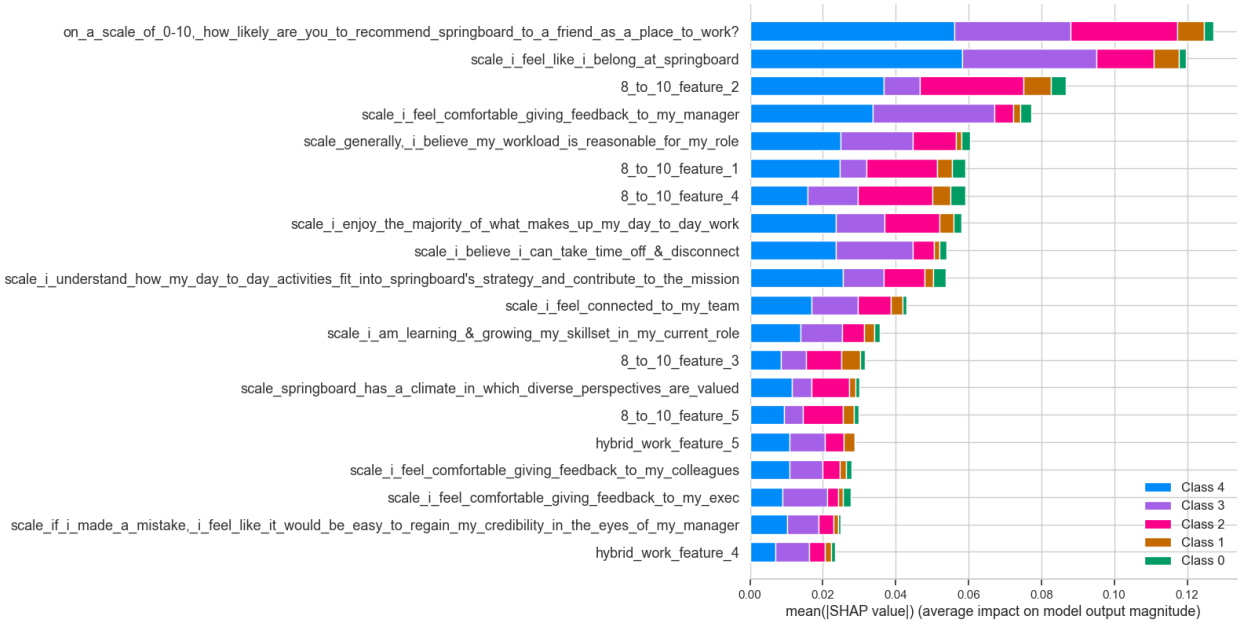


Exhibit 7: True Positive and False Positive Rates of Model 2 by Percentile

Percentile	Num_Employees	RF_TPR	RF_FPR	RF_Precision	GBM_TPR	GBM_FPR	GBM_Precision	XGB_TPR	XGB_FPR	XGB_Precision	CAT_TPR	CAT_FPR	CAT_Precision
20.0	36	1.000000	1.000000	0.888889	1.000000	1.000000	0.833333	1.000000	1.000000	0.777778	1.0	1.0	0.916667
40.0	72	1.000000	1.000000	0.833333	1.000000	1.000000	0.833333	1.000000	1.000000	0.791667	1.0	1.0	0.861111
60.0	108	1.000000	1.000000	0.796296	1.000000	1.000000	0.759259	1.000000	1.000000	0.740741	1.0	1.0	0.805556
80.0	144	0.963303	0.685714	0.813953	0.941176	0.785714	0.744186	0.939394	0.755556	0.732283	0.943396	0.684211	0.793651
100.0	180	0.882353	0.393443	0.813953	0.806723	0.540984	0.744186	0.781513	0.557377	0.732283	0.840336	0.42623	0.793651

Note: Percentiles are based on the result of ranking each data point by its predicted value. For each data point, a probability is assigned to each class (1-5) by the model, with the sum of these probabilities adding up to 1.0. The "score" we use to rank the data points is given by P(class 1) × 1 + P(class 2) × 2 + … + P(class 5) × 5.

The table above is for Model 2, which uses the original dataset (180 employee responses) along with sentiment analysis features. The 20th percentile corresponds to the top 36 employees, the 40th percentile corresponds to the top 72 employees, etc. until we eventually cover all 180 employees at the 100th percentile.

Appendix

The following is a link to a Google Drive folder with our code: Springboard Project Code
(Note: must be logged in to a Stanford Google account to view)

This folder contains the code used to clean our data, run Models 1-3, and generate our figures. The following is a description of each file. There are 3 notebook files and 10 data files, for a total of 13 files.

Notebooks:

1. Springboard_Final_Workflow.ipynb: Code for training Models 1 and 2
2. Springboard_Final_Workflow_Extended_Data.ipynb: Code for training Model 3
3. Sentiment_FULL.ipynb: Code to generate sentiment analysis features and perform PCA

The data_files folder contains all of the input data used in the above notebooks. It also contains output files that are generated from the above notebooks.

Data Files:

- Springboard Pulse Survey (May 2022) (Responses).xlsx: Original dataset provided by Springboard with 180 employee responses.
- springboard_data_cleaned.csv: Output file from Notebook 1. Contains cleaned data used for training Models 1 and 2.
- SpringboardPulseBIG.xlsx: Springboard dataset augmented with January 2022 and February 2023 survey data.
- SpringboardPulseBIGBinned.csv: Augmented Springboard dataset (see above), with binned (one-hot encoding) categorical features. Output file from Notebook 2 that is later read in by the same notebook to train Model 3 ("Three Survey Model with Sentiment and PCA Analysis").
- FreeForm_BIG.xlsx: Only the freeform survey question columns from the SpringboardPulseBIG.xlsx file. Used to obtain the sentiment analysis features below.
- Sentiment analysis features:
    - df_sentiment.csv: Sentiment analysis features (each with a value ranging from 1-5) from only a basic BERT model on all four freeform survey questions.
    - Feat_REAL_BERT_FreeForm_8_to_10.csv: Sentiment analysis features (obtained from BERT followed by PCA) for the following freeform survey question: "If your score is anything between 0-8, what would it take to move your rating to 9 or 10"?
    - Feat_REAL_BERT_FreeForm_additional_feedback.csv: Sentiment analysis features for the freeform survey question: "Do you have any additional feedback to share about your organization and/or about Springboard?"

- Feat_REAL_BERT_FreeForm_cilantro.csv: Sentiment analysis features for the following freeform survey question: “What do you think is Springboard’s ‘cilantro’ (something Springboarders love but others might dislike)?”
- Feat_REAL_BERT_FreeForm_hybrid_work.csv: Sentiment analysis features for the freeform survey question: “What changes would make you feel better about Springboard’s approach to hybrid work?”